

Lab: Securing a SOHO Network

Estimated time needed: **20** minutes.

Objectives

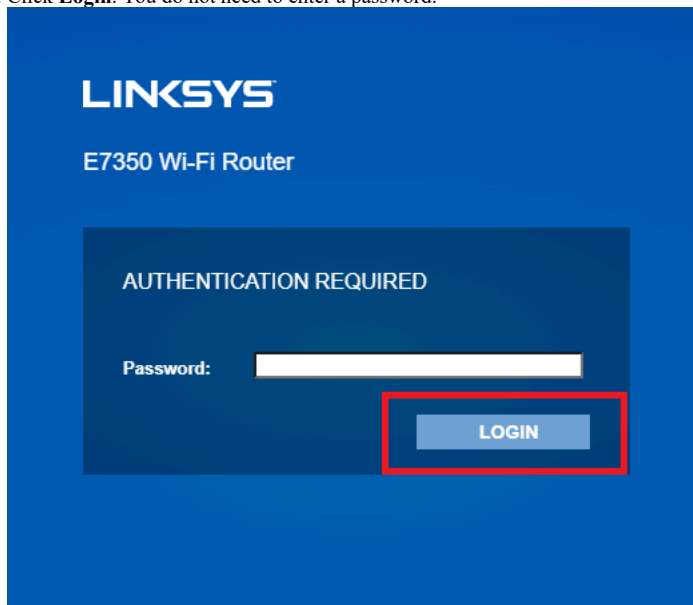
- Change the default administrator password of the SOHO router to enhance security
- Change the SOHO router's default Service Set Identifier (SSID) for improved wireless security
- Enable the WPA3 Personal security protocol with a strong passphrase for the SOHO router's 2.4 GHz and 5 GHz frequencies

Exercise 1: Change the Administrator Router Password

In this exercise, you will change the administrator password on a router.

Steps

1. Access the Cisco Linksys E7350 emulator using the following URL: <https://ui.linksys.com/ExpertFamily/E7350/>
2. Click **Login**. You do not need to enter a password.



3. Select the **Configuration** tab.

LINKSYS

AX1800 Wi-Fi R

System Status

Configuration

Parental Controls

Troubleshoot

Status

Router

Local Network

Wireless Network

Ports

Router

Router Information

Firmware Version

1.1.00 build 9 Jan 26, 2021

Current Time

Tue,07 Dec 2021 11:22:26

Internet MAC Address

30:23:03:06:93:84

Router Name

Linksys04253

Host Name

Not Available

Domain Name

Not Available

Internet Connection

IPv4

Connection Type

Automatic Configuration - DHCP

Internet IP Address

172.30.1.125

Subnet Mask

255.255.255.0

Default Gateway

172.30.1.1

DNS1

172.30.1.1

DNS2

Not Available

DNS3

Not Available

MTU

1500

DHCP Lease Time

24 Hour

Release IP Address

Renew IP Address

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4. Click **Administration** in the left panel.

LINKSYS

AX1800 Wi-Fi 6 Router

System Status

Configuration

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Troubleshooting

Connectivity

Local Network

WAN Setup

VLAN Setup

IPv6 Setup

MAC Address Clone

Advanced Routing

Wi-Fi

External Storage

Security

Administration

Local Network

Router Address

IP Address19216811

Subnet Mask255.255.255.0

Router NameLinksys04253

DHCP Server Setting

☒ DHCP Server

DHCP Reservation

Start IP Address1921681100

Maximum Number of Users150

IP Address Range1921681100 to 192.168.1.249

Client Lease Time1440 minutes (0 means one day)

Static DNS 10000

Static DNS 20000

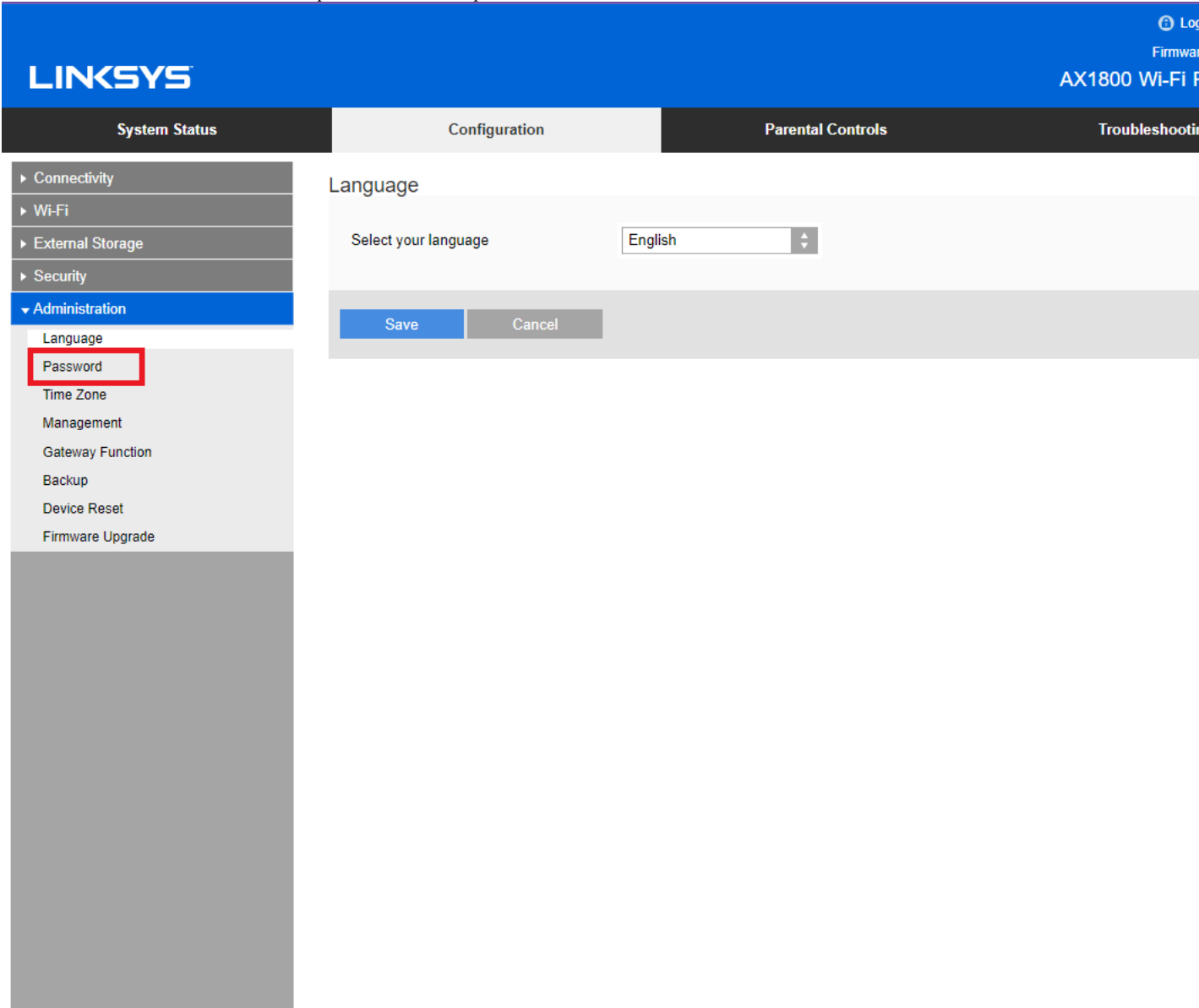
Static DNS 30000

WINS0000

Save

Cancel

5. Select **Password** in the Administration sub-panel to create a new password.



6. For creating a password, enter at least 12 characters, including a mix of uppercase and lowercase letters, numbers, and special characters. Avoid using common words, personal information, or easily guessable patterns in the password. For this example, type **K1b2!79100@1** into the **Router password** and the **Re-Enter to**

Confirm boxes.

LINKSYS

AX1800 Wi-Fi R

System Status

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Connectivity

Wi-Fi

External Storage

Security

Administration

Language

Password

Time Zone

Management

Gateway Function

Backup

Device Reset

Firmware Upgrade

Log

Firmwa

Router Password

Re-Enter to Confirm

Save

Cancel

7. Click **Save** to continue.

LINKSYS

AX1800 Wi-Fi R

System Status

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External Storage

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Administration

Language

Password

Time Zone

Management

Gateway Function

Backup

Device Reset

Firmware Upgrade

Password

Router Password

Re-Enter to Confirm

Save

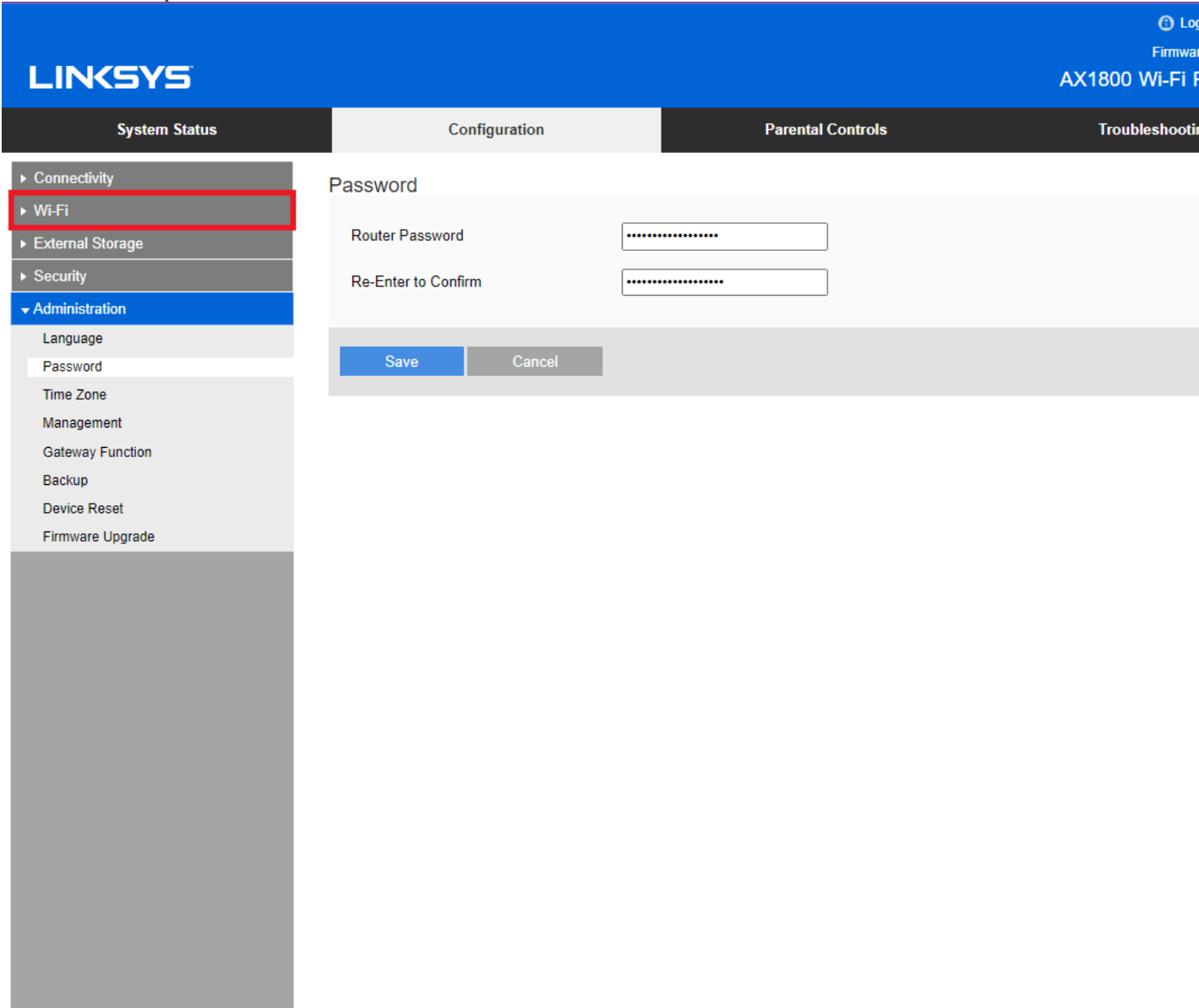
Cancel

Exercise 2: Change the Router's Default SSID

In this exercise, you will change the router's default network name (SSID).

Steps

1. Select **Wi-Fi** in the left panel.



2. Select **Basic Wireless Settings** in the Wi-Fi sub-panel. Create a custom SSID on the **Basic Wireless Settings** screen. Avoid using personal information such as a name or address. Unless needed for guest access, your SSID should not be associated with the location or purpose of the network. For this example, type **my wi-fi 5**

into the **Network Name (SSID)** under the **5 Ghz Wireless Settings**.

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AX1800 Wi-Fi R

System Status

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Wi-Fi

Basic Wireless Settings

Wi-Fi Protected Setup

Wireless Security

Guest Access

Wireless MAC Filter

Advanced Settings

External Storage

Security

Administration

Basic Wireless Settings

EasyMesh

Add a new node

Mesh information

5 GHz Wireless Settings

Network Mode

Mixed(A/N/AC/AX)

Network Name (SSID)

Linksys_5Ghz

Channel Width

Auto

Channel

Auto

SSID Broadcast

2.4 GHz Wireless Settings

Network Mode

Mixed(B/G/N/AX)

Network Name (SSID)

Linksys_2.4Ghz

Channel Width

Auto

Channel

Auto

SSID Broadcast

Save

Cancel

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3. Type **my wi-fi 2.4** into the **Network Name (SSID)** under the **2.4 Ghz Wireless Settings**.

LINKSYS

AX1800 Wi-Fi R

System Status

Configuration

Parental Controls

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Connectivity

Wi-Fi

Basic Wireless Settings

Wi-Fi Protected Setup

Wireless Security

Guest Access

Wireless MAC Filter

Advanced Settings

External Storage

Security

Administration

Basic Wireless Settings

EasyMesh

Add a new node

Mesh information

5 GHz Wireless Settings

Network Mode

Mixed(A/N/AC/AX)

Network Name (SSID)

my wi-fi 5

Channel Width

Auto

Channel

Auto

SSID Broadcast

2.4 GHz Wireless Settings

Network Mode

Mixed(B/G/N/AX)

Network Name (SSID)

Linksys_2.4Ghz

Channel Width

Auto

Channel

Auto

SSID Broadcast

Save

Cancel

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4. Click **Save** to apply the new SSIDs.

LINKSYS

AX1800 Wi-Fi R

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Basic Wireless Settings

Wi-Fi Protected Setup

Wireless Security

Guest Access

Wireless MAC Filter

Advanced Settings

External Storage

Security

Administration

Basic Wireless Settings

☒ EasyMesh

Add a new node

Mesh information

5 GHz Wireless Settings

Network Mode

Mixed(A/N/AC/AX)

Network Name (SSID)

my wi-fi 5

Channel Width

Auto

Channel

Auto

☒ SSID Broadcast

2.4 GHz Wireless Settings

Network Mode

Mixed(B/G/N/AX)

Network Name (SSID)

my wi-fi 2.4

Channel Width

Auto

Channel

Auto

☒ SSID Broadcast

Save

Cancel

Exercise 3: Enable WPA3 Personal with Passphrase for Both Frequencies

In this exercise, you will enable WPA3 security so only those with the correct passphrase can connect their device to the wireless network.

Steps

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1. Click **Wireless Security** under the **Wi-Fi** menu.

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AX1800 Wi-Fi R

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Wi-Fi

Basic Wireless Settings

Wi-Fi Protected Setup

Wireless Security

Guest Access

Wireless MAC Filter

Advanced Settings

External Storage

Security

Administration

Basic Wireless Settings

EasyMesh

Add a new node

Mesh information

5 GHz Wireless Settings

Network Mode

Mixed(A/N/AC/AX)

Network Name (SSID)

my wi-fi 5

Channel Width

Auto

Channel

Auto

SSID Broadcast

2.4 GHz Wireless Settings

Network Mode

Mixed(B/G/N/AX)

Network Name (SSID)

my wi-fi 2.4

Channel Width

Auto

Channel

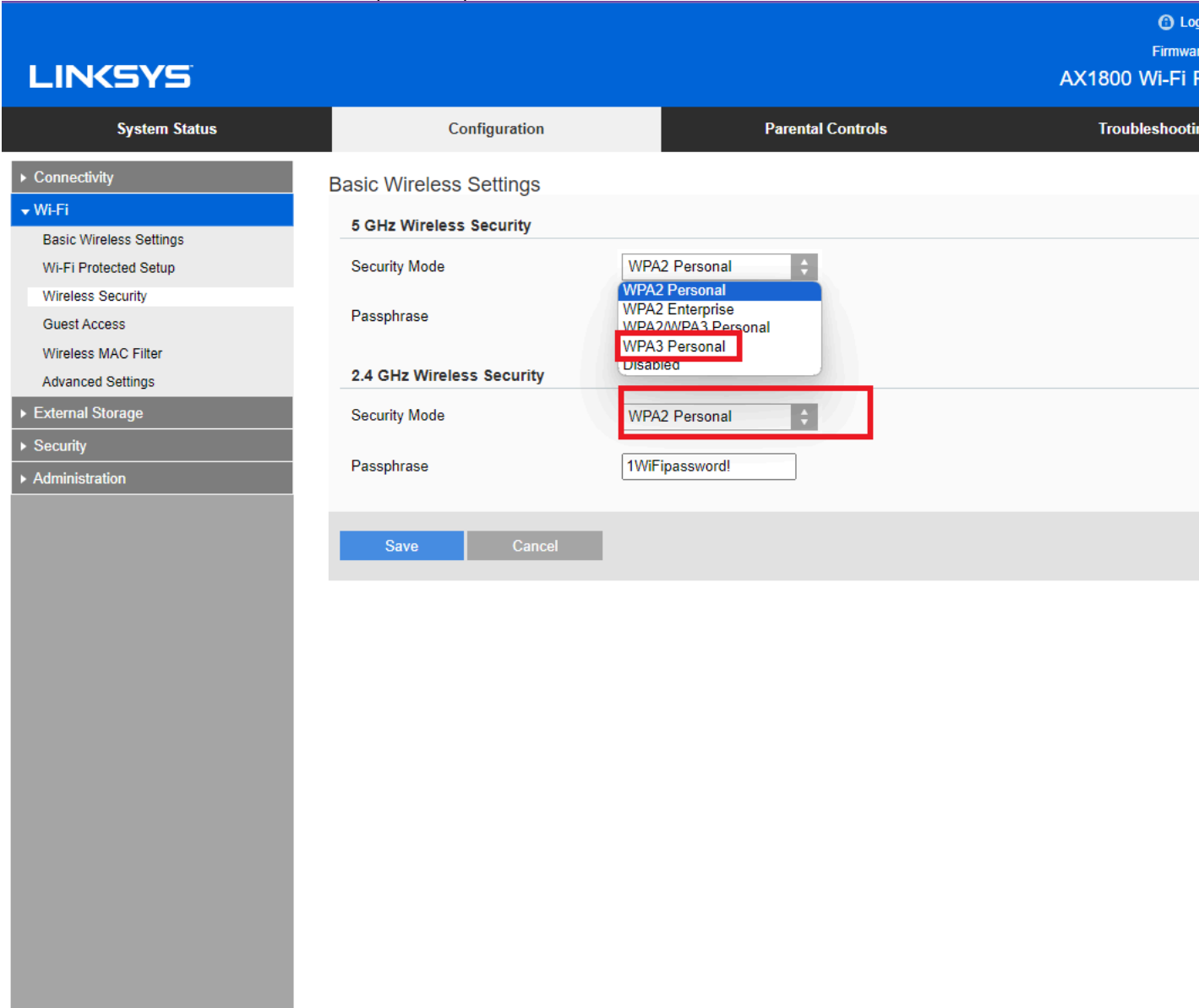
Auto

SSID Broadcast

Save

Cancel

2. Select **WPA3 Personal** from the 2.4 and 5 GHz frequencies dropdown menu.



3. Enter a strong passphrase that is at least 12 characters long. For this example, use **C0mp13xP@ss!** Enter the same password for the 2.4 and 5 GHz **Passphrase**.

LINKSYS

AX1800 Wi-Fi R

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Connectivity

Wi-Fi

Basic Wireless Settings

Wi-Fi Protected Setup

Wireless Security

Guest Access

Wireless MAC Filter

Advanced Settings

External Storage

Security

Administration

Basic Wireless Settings

5 GHz Wireless Security

Security Mode

WPA3 Personal

Passphrase

1WiFipassword!

2.4 GHz Wireless Security

Security Mode

WPA3 Personal

Passphrase

1WiFipassword!

Save

Cancel

4. Click **Save** to enable WPA3 security with the specified passphrase.

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AX1800 Wi-Fi R

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Wi-Fi

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Wi-Fi Protected Setup

Wireless Security

Guest Access

Wireless MAC Filter

Advanced Settings

External Storage

Security

Administration

Basic Wireless Settings

5 GHz Wireless Security

Security Mode

WPA3 Personal

Passphrase

C0mp13xP@ss!

2.4 GHz Wireless Security

Security Mode

WPA3 Personal

Passphrase

C0mp13xP@ss!

Save

Cancel

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Skills Network